

The density expansion of a face term

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Abstract

For a one-variable amplitude $\Psi(u, s)$ with u -Taylor data $f_m(s)$ to order M and weights $u^{h_1}v^{h_2}$, $p_2 = (h_2 + 1)/k_2$, the density of unit 103 is expanded by the weighted finite-part axis lemma (unit 100) with $\gamma = k_1 p_2 - h_1 - 1 < M$:

$$\rho(r, s) = \sum_{m < M} r^{p_2 + \frac{m-\gamma}{k_1} - 1} c_m(s) + r^{p_2 - 1} k_2^{-1} \text{FP}_\gamma \Psi(\cdot, s) + r^{p_2 - 1} \log \frac{1}{r} \frac{f_\gamma(s)}{k_1 k_2} + O(r^{p_2 + \frac{M-\gamma}{k_1} - 1} H(s)),$$

with the logarithm present exactly when γ is an integer below M (a collision of exponents) and explicit c_m . The expansion is packaged as a `DensityExpansion` indexed by $\text{Fin } M \oplus \text{Fin } 2$ and transferred to the N -expansion of the face term. This is the first concrete instance of Astra #4's cutoff-indexed $d = 2$ density theorem: one face term, all of its exponents, its resonant logarithm and its finite-part coefficient. Zero sorry/axiom.

1 The things formalised

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noncomputable def faceExp (p : ℕ) (k : ℕ) (m : ℕ) : ℝ := p + ((m : ℕ) - 1) / k
noncomputable def faceCoeff (A : ℝ) (k k : ℕ) (f : ℝ → ℝ) (m : ℕ) (s : ℝ) : ℝ :=
  if (m : ℕ) = 0 then 1 / ((k : ℕ) * k) * Real.log A * f m s
  else -(1 / (k : ℕ)) * A ^ (-((m : ℕ) - 1) / k) / ((m : ℕ) - 1) * f m s
noncomputable def faceLogCoeff (A : ℝ) (k k : ℕ) (f : ℝ → ℝ) (M : ℕ) (s : ℝ) : ℝ :=
  m range M, if (m : ℕ) = 0 then 1 / ((k : ℕ) * k) * f m s else 0
noncomputable def faceFPCoeff (b : ℝ) (k : ℕ) (Ψ : ℝ → ℝ) (f : ℝ → ℝ) (M : ℕ)
  (s : ℝ) : ℝ := 1 / (k : ℕ) * axisFinitePart b (fun u => Ψ u s) (fun m => f m s) M
noncomputable def face / def faceJ / noncomputable def faceC
  -- indexed by Fin M Fin 2
noncomputable def faceEnv (A : ℝ) (k k M : ℕ) (H : ℝ → ℝ) (s : ℝ) : ℝ :=
  1 / (k : ℕ) * (A ^ (-((M : ℕ) - 1) / k)) / ((M : ℕ) - 1) * H s
theorem face_monomial_eq ... :
  -(1 / (k : ℕ)) * r ^ (p - 1) * (f m s * axisPrim ((r / b ^ k) ^ ((k : ℕ) ^ 1)) m)
  = r ^ (faceExp p k m - 1) * faceCoeff (b ^ k) k k f m s
  + r ^ (p - 1) * Real.log (1 / r)
  * (if (m : ℕ) = 0 then 1 / ((k : ℕ) * k) * f m s else 0)
theorem face_terms_eq ... -- local expression = r^{-1} log(1/r)^{j} c(s)
theorem genDensity_measurable / faceFPCoeff_measurable / faceC_measurable
theorem faceDensity_expansion (p b : ℝ) (h k k M : ℕ) (hb : 0 < b) (hk : 0 < k)
  (Ψ : ℝ → ℝ) (hΨc : Continuous (Function.uncurry Ψ)) (f : ℝ → ℝ)
  (hf : m, Measurable (f m)) (H : ℝ → ℝ) (hH : Measurable H) (hH0 : s, 0 < H s)
  (h : (k : ℕ) * p - h - 1 < M)
  (hrem : s, 0 < s → u Ioc (0 : ℝ) b,
    |Ψ u s - m range M, f m s * u ^ m| H s * u ^ M) :

```

```

DensityExpansion (genDensity p b h k k Ψ) (face p k M) (faceJ M)
  (faceC b k k Ψ f M) (p + ((M : ) - ) / k) (b ^ (k + k)) 1
  (faceEnv (b ^ k) k k M H) -- = k p -h -1
theorem face_transfer_bound ... (hp : ((h : ) + 1) / k = p) ... (N : ) (hN : 1 N) :
  |twoDamp b N h h k k (fun u _ s => Ψ u s)
   - i, transferTerm (face ..) (faceJ ..) (faceC ..) N|
  N ^ (-(p + (M - )/k)) * (1 + Real.log N) ^ 1
  * (envMoment .. + i, R ^ ( - a) * tailMoment ..)
theorem face_transfer_sum_eq ... :
  i, transferTerm (face p k M i) (faceJ M i) (faceC b k k Ψ f M i) N
  = ( m : Fin M, N ^ (-(faceExp p k m))
    * logMoment (faceExp p k m) 0 (faceCoeff (b ^ k) k k f m))
  + N ^ (-p) * logMoment p 0 (faceFPCoeff b k Ψ f M)
  + N ^ (-p) * (Real.log N * logMoment p 0 (faceLogCoeff k k f M)
    - logMoment p 1 (faceLogCoeff k k f M))

```

2 Mathematics

With $\varepsilon(r) = (r/b^{k_2})^{1/k_1}$ and $A = b^{k_2}$, unit 100 gives $J(\varepsilon) = \text{FP}_\gamma - \sum_{m < M} f_m \text{axisPrim}(\gamma, \varepsilon, m) + E$ with $|E| \leq H\varepsilon^{M-\gamma}/(M-\gamma)$. The monomial terms are converted to powers of r by `axisPrim_rpow`: off resonance $(r/A)^{(m-\gamma)/k_1}/(m-\gamma) = r^{(m-\gamma)/k_1} A^{-(m-\gamma)/k_1}/(m-\gamma)$, so the term has exponent $p_2 + (m-\gamma)/k_1$; at the resonance $m = \gamma$, $(\log r - \log A)/k_1$ splits into a $\log(1/r)$ term with coefficient $(k_1 k_2)^{-1} f_\gamma$ and a constant $(k_1 k_2)^{-1} \log A f_\gamma$ at exponent p_2 (`face_monomial_eq`). The remainder $k_2^{-1} r^{p_2-1} \varepsilon^{M-\gamma} H/(M-\gamma)$ has exponent $a = p_2 + (M-\gamma)/k_1$, strictly above every term since $m < M$; the uniform log degree 1 absorbs the resonant term. Measurability of the finite-part coefficient in s is a parametric-integral argument (`StronglyMeasurable.integral_prod_right'`), and of the density in (r, s) the indicator trick of unit 87.

3 Paper-facing meaning

A face term $u^i a_i(v, s)$ of the rectangular decomposition (after swapping coordinates) is exactly this situation with $\gamma = k_2 i/k_1$, so its contribution to the $d = 2$ block is

$$\begin{aligned}
& \sum_{m < M} N^{-(p+m/k_2)} \mathcal{M}_{p+m/k_2, 0}(c_m) + N^{-(p+i/k_1)} \left[\mathcal{M}_{p+i/k_1, 0}(k_2^{-1} \text{FP}) \right. \\
& \left. + \log N \mathcal{M}_{p+i/k_1, 0}(c_{\log}) - \mathcal{M}_{p+i/k_1, 1}(c_{\log}) \right] + O(N^{-(p+M/k_2)}(1 + \log N)),
\end{aligned}$$

with $c_{\log} \neq 0$ only when $i/k_1 = m/k_2$ for some $m < M$: the logarithms of the higher-order $d = 2$ expansion occur precisely at collisions of the two exponent lattices $p + \mathbb{N}/k_1$ and $p + \mathbb{N}/k_2$, as predicted in Astra #3(b)/#4(b). What remains for the full τ -cutoff theorem is the rectangular decomposition $\Phi = \sum_i u^i a_i + \sum_j v^j b_j - \sum_{ij} c_{ij} u^i v^j + R$ with the mixed remainder estimate, and the bookkeeping of the corner monomials.

4 Lean notes

- A file's import closure need not contain everything below it in `Laplace.lean`: `axisFinitePart` was unknown until `AxisSeries` (hence `FinitePartAxis`) was imported explicitly.

- `Fintype.sum_sum_type` + `Fin.sum_univ_two` + `simp only [Sum.elim_inl, Sum.elim_inr, Matrix.cons_val_zero, Matrix.cons_val_one]` unfold a `Fin M` `Fin` 2-indexed sum; `Fin.sum_univ_eq_sum_range` aligns the `Fin M` sum with a `range M` sum; the final step is `linear_combination (-1 : ) * hS`.
- `simp only` may already distribute a scalar over a sum, so a later `rw [Finset.mul_sum]` finds nothing; prove the combined identity as a `have` instead.
- A `gcongr` side goal `0 faceEnv ...` is reported as an unsolved `case ha` at the enclosing `have`; opaque definitions need the monotonicity lemma spelled out (`mul_le_mul_of_nonneg_right`).
- `split_ifs` on a condition independent of the variable `((m : ) = )` is the cheap way to get measurability of an `ite`-defined coefficient.